NMAM Institute of Technology, Nitte

Off-Campus Centre of Nitte (Deemed to be University)

COMPREHENSIVE EXAMINATION QUESTION BANK

IV Semester B.E./B.Tech (CBCS)

Academic Years: 2022–2025

MSE-I, MSE-II, SEE & Supplementary Examinations

Subjects Included:

Part A:	MCQs + Descriptive (From Images)
MA2005-1	Linear Algebra and Numerical Methods
CS2102-1	Database Management Systems
CS3004-1	Design and Analysis of Algorithms
ME1654-1	Innovation and Design Thinking
21CS403	Microprocessor and Microcontrollers
21CS404	Software Engineering & Project Management
Part B:	Descriptive Only (From PDF Source)
CS2103-1/AD1103-1	Software Engineering & Project Management
CS3005-1	Microprocessor and Embedded Systems
CS2101-1	Computer Organization and Architecture

Complete Collection – All Papers Merged

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Part I
MCQs + Descriptive Questions (From Images)

1 Linear Algebra and Numerical Methods (MA2005-1)

1.1 MSE-I – February 2025

NMAM INSTITUTE OF TECHNOLOGY, NITTE

Off-Campus Centre of Nitte (Deemed to be University)

IV SEM B.Tech (CBCS) Mid Semester Examinations - I, February 2025

MA2005-1 – LINEAR ALGEBRA AND NUMERICAL METHODS

(For AD, AM, CB, CC, CS, IS)

Duration: 1 Hour Max. Marks: 15

Part – A: Multiple Choice Questions ($3 \times 1 = 3$ marks)

1. For what values of k will the augmented matrix $\left[\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 2 & 2 & 1 & 3 \\ 1 & 2 & k & k+1 \end{array}\right]$ correspond to an inconsistent system of linear equations?
   (A) $k \neq 1$ (B) $k = 0$ (C) $k = 1$ (D) All values of k

2. If $A = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 \\ 1 & 2 \end{bmatrix}$, find $(AB)^{-1}$.
   (A) $\begin{bmatrix} 1 & 1/3 \\ -1/6 & 1/2 \end{bmatrix}$ (B) $\begin{bmatrix} 1 & -1/2 \\ -1/6 & 1/2 \end{bmatrix}$ (C) $\begin{bmatrix} 1 & -1/3 \\ -1/6 & 1/2 \end{bmatrix}$ (D) $\begin{bmatrix} 1 & -1/3 \\ 1/6 & 1/2 \end{bmatrix}$

3. Given a matrix A, column 1 and column 3 are pivot columns, then the dimension of Col A is:
   (A) 1 (B) 2 (C) 3 (D) 4

Part – B: Descriptive Answer Questions ($2 \times 6 = 12$ marks)

Unit – I

1. a) Find the eigenvalues of $A = \begin{bmatrix} 1 & 2 & 1 \\ 6 & -1 & 0 \\ -1 & -2 & -1 \end{bmatrix}$. Also determine the eigenvectors corresponding to each eigenvalue. [6 marks]

   b) Reduce the matrix $A = \begin{bmatrix} -2 & -4 & 2 \\ -2 & 1 & 2 \\ 4 & 2 & 5 \end{bmatrix}$ to the form $PA = LU$, using partial pivoting. [6 marks]

Unit – II

2. a) Solve by Gauss-Seidel method: $10x + y + z = 12$; $2x + 10y + z = 13$; $x + y + 5z = 7$. (3 iterations) [6 marks]

   b) Find the dominant eigenvalue and eigenvector of $A = \begin{bmatrix} 1 & 6 & 1 \\ 1 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$ using Rayleigh's Power method (4 iterations). [6 marks]

1.2 MSE-II – April 2025

NMAM INSTITUTE OF TECHNOLOGY, NITTE

Off-Campus Centre of Nitte (Deemed to be University)

IV SEM B.Tech (CBCS) Mid Semester Examinations - II, April 2025

MA2005-1 – LINEAR ALGEBRA AND NUMERICAL METHODS

(For AD, AM, CB, CC, CS, IS)

Duration: 1 Hour Max. Marks: 20

Part – A: Multiple Choice Questions ($4 \times 1 = 4$ marks)

1. Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be a linear transformation such that $T(x_1, x_2) = (x_1+x_2, x_1)$. Then $T^{-1}(x_1, x_2)$ equals:
   (A) $(x_2, x_1 - x_2)$ (B) $(x_2, x_1 + x_2)$ (C) $(x_1, x_1 - x_2)$ (D) $(x_1, x_1 + x_2)$

2. The set $\{1, x, x^2\}$ forms a basis for:
   (A) P_1 (B) P_2 (C) P_3 (D) P_4

3. Which is a subspace of the given vector space?
   (A) Set of all polynomials of degree exactly 3, in P_3
   (B) Set of all polynomials with positive coefficients, in P
   (C) Set of all 2×2 matrices with positive entries
   (D) Set of all symmetric 2×2 matrices, in $M_{2\times 2}$

4. Given matrix $A_{4\times 5}$ with $\text{rank}(A) = 3$, then $\dim(\text{Null}(A))$ is:
   (A) 1 (B) 2 (C) 3 (D) 4

Part – B: Descriptive ($2 \times 8 = 16$ marks)

Unit – I

1. a) Let $T : P_2 \to P_2$ be the transformation $p(t) \mapsto p(t + 2)$. Find image of $p(t) = 3 - 2t + t^2$. Show T is linear. Find matrix with basis $B = \{1, t, t^2\}$. [8 marks]

   b) Let $\vec{v_1} = (1, -2, 3)$, $\vec{v_2} = (5, 6, -1)$, $\vec{v_3} = (3, 2, 1)$. Show $S = \{\vec{v_1}, \vec{v_2}, \vec{v_3}\}$ is a basis for $\mathbb{R}^3$. [8 marks]

Unit – II

2. a) Use Gram-Schmidt to construct orthonormal basis from $B = \{(1, 1, 1), (0, 1, 1), (0, 0, 1)\}$. Find QR factorization. [8 marks]

   b) Use coordinate vectors to test linear independence of $S = \{1 - 2t^2 - t^3, t + 2t^3, 1 + t - 2t^2\}$. [8 marks]

1.3 SEE – May 2025

NMAM INSTITUTE OF TECHNOLOGY, NITTE

Off-Campus Centre of Nitte (Deemed to be University)

Fourth Semester B.Tech (CBCS) Degree Examinations, May 2025

MA2005-1 – LINEAR ALGEBRA AND NUMERICAL METHODS

(For AD, AM, CB, CC, CS, IS)

Duration: 3 Hours Max. Marks: 100

Part – A: MCQs (20 × 1 = 20 marks)

1. For what value of k, the augmented matrix corresponds to inconsistent system?
$\begin{bmatrix} 1 & 3 & -2 & | & 3 \\ 3 & k & -6 & | & 9 \\ 1 & -3 & 2 & | & 5 \end{bmatrix}$
(A) $k = 1$ (B) $k = 9$ (C) $k = -3$ (D) None

2. If A and B are matrices such that AB exists and is square, then:
(A) Both are square of same order (B) Number of rows of A = columns of B
(C) BA need not exist (D) One of them is identity

3. Columns 1, 2, 5 are pivot columns. Which is NOT necessarily true?
(A) dim(Col A) = 3 (B) dim(Row A) = 3 (C) rank(A) = 3 (D) dim(Null A) = 3

4. If λ is eigenvalue of A, eigenvalue of A^{-1} is: (A) λ (B) $-\lambda$ (C) λ^2 (D) $1/\lambda$

5. If $A = \begin{bmatrix} 1 & 3 & 5 \\ 2 & 3 & 1 \\ 0 & 0 & 2 \end{bmatrix}$ then $A^3 - 3A^2 - A + 3I =$ (A) $A + I$ (B) I (C) $2A$ (D) 0

6. Eigenvalues of $P^{-1}AP$ are same as: (A) PA (B) A (C) P (D) Cannot say

7. Gauss-Seidel: arrange so that diagonal elements are: (A) Any order (B) Zero (C) Larger than others (D) Cannot solve

8. Which is a direct method? (A) Jacobi (B) Gauss-Seidel (C) LU Decomposition (D) Power method

9. If λ is eigenvalue of A, then $\lambda^2 + 1$ is eigenvalue of: (A) $A^2 + I$ (B) $A^2 + A$ (C) $2A$ (D) $(A + I)^2$

10. If $1 + x$, $1 + x + x^2$, $ax^2 + x^3$ are linearly dependent, $a =$ (A) 1 (B) 2 (C) 3 (D) 4

11. Dimension of $W = \{(a, b, c, d) : a - 2b + c - d = 0\}$ is: (A) 1 (B) 2 (C) 3 (D) 4

12. V = set of 2×2 symmetric matrices, dimension is: (A) 1 (B) 2 (C) 3 (D) 4

13. For $T : V \to W$, $T^{-1}(U)$ is subspace of: (A) V (B) W (C) Both (D) Neither

14. $T : \mathbb{R}^n \to \mathbb{R}^m$ can be matrix multiplication: (A) Only if $m = n$ (B) Only if $m \neq n$ (C) Always (D) Never

15. If $\vec{u}, \vec{v}$ orthogonal: $\|\vec{u} + \vec{v}\|^2 =$ (A) $\|\vec{u}\|^2 + \|\vec{v}\|^2$ (B) $\|\vec{u}\|^2 - \|\vec{v}\|^2$ (C) Product (D) Quotient

16. $\vec{u} = (1, 2, -3)$, $\vec{v} = (a, 1, 1)$ orthogonal if $a =$ (A) 1 (B) 2 (C) 3 (D) 5

17. Linear combination of rows of A: (A) $\vec{x}^T A$ (B) $A\vec{x}$ (C) $\vec{x}A^T$ (D) $A^T\vec{x}$

18. Angle between $\vec{u} = (1, 2, -1, 0)$ and $\vec{v} = (-1, 1, 1, 2)$: (A) $\pi/4$ (B) $\pi/3$ (C) $\pi/2$ (D) $2\pi/3$

19. Which defines inner product on $\mathbb{R}^2$? (A) $u_1v_1 - u_2v_2$ (B) $u_1v_1 + 2u_2v_2$ (C) $u_1^2v_1^2 + u_2^2v_2^2$ (D) $u_1v_2 - u_2v_1$

20. For orthonormal basis: $\|\vec{x}\|^2 =$ (A) $\sum |\langle \vec{x}, \vec{u}_i \rangle|^2$ (B) $\sum \langle \vec{x}, \vec{u}_i \rangle$ (C) $\sum |\langle \vec{x}, \vec{u}_i \rangle|$ (D) None

Part – B: Descriptive (80 marks)

[Questions 1-6 with eigenvalues, LU factorization, Gauss-Seidel, Power method, vector spaces, linear transformations, Gram-Schmidt, QR factorization, SVD, least squares - each 16 marks]

2 Database Management Systems (CS2102-1)

2.1 MSE-I – February 2025

NMAM INSTITUTE OF TECHNOLOGY, NITTE

Off-Campus Centre of Nitte (Deemed to be University)

IV SEM B.Tech (CBCS) Mid Semester Examinations - I, February 2025

CS2102-1 – DATABASE MANAGEMENT SYSTEMS

(For AD, AM, CB, CC, CS, IS)

Duration: 1 Hour Max. Marks: 15

Part – A: MCQs ($3 \times 1 = 3$ marks)

1. A ____ is a unit of storage containing records: (A) Block (B) Field (C) Tuple (D) File
2. NOT a component of ER Model: (A) Entity (B) Attribute (C) Table (D) Relationship
3. Used to uniquely identify a record: (A) Foreign Key (B) Candidate Key (C) Primary Key (D) Super Key

Part – B: Descriptive (12 marks)

1. a) Draw ER diagram for University Database with Student, Course, Instructor, Department. [6 marks]

   b) Explain three-schema architecture of DBMS with diagram. [6 marks]
2. a) Movie database relational algebra queries. [6 marks]

   b) Explain different types of keys in DBMS with examples. [6 marks]

2.2 MSE-II – April 2025

NMAM INSTITUTE OF TECHNOLOGY, NITTE

Off-Campus Centre of Nitte (Deemed to be University)

IV SEM B.Tech (CBCS) Mid Semester Examinations - II, April 2025

CS2102-1 – DATABASE MANAGEMENT SYSTEMS

(For AD, AM, CB, CC, CS, IS)

Duration: 1 Hour Max. Marks: 20

Part – A: MCQs ($4 \times 1 = 4$ marks)

1. "No component of Primary Key accepts Null" is: (A) Referential Integrity (B) Domain Integrity (C) Entity Integrity (D) Tuple Integrity
2. Add new column to table: (A) ALTER (B) UPDATE (C) INSERT (D) MODIFY
3. BCNF: every non-trivial FD $X \to Y$: (A) X is super key (B) Y is prime (C) X is candidate key (D) Y is super key
4. Combine tuples based on condition: (A) Union (B) Join (C) Projection (D) Selection

Part – B: Descriptive (16 marks)

Suppliers-Parts-Catalog SQL Queries:

```
Suppliers(sid, sname, address)
Parts(pid, pname, color)
Catalog(sid, pid, cost)
```

i) Find names of suppliers who supply some red part.

ii) Find sids of suppliers who supply some red or green part.

iii) Find sids who supply red part or are at 221 Packer street.

iv) Find pids supplied by at least two different suppliers.

Functional Dependencies: $R(A, B, C, D, E)$ with FDs: $\{A \to BC, CD \to E, B \to D, E \to A\}$. Find candidate keys. Check equivalence.

Normalization: $R(A, B, C, D)$ with $\{A \to B, B \to C, C \to D\}$. Find candidate keys. Check 3NF. Decompose if needed.

2.3 SEE – May 2025

NMAM INSTITUTE OF TECHNOLOGY, NITTE

Off-Campus Centre of Nitte (Deemed to be University)

Fourth Semester B.Tech (CBCS) Degree Examinations, May 2025

CS2102-1 – DATABASE MANAGEMENT SYSTEMS

(For AD, AM, CB, CC, CS, IS)

Duration: 3 Hours Max. Marks: 100

Part – A: MCQs (20 marks)

[20 MCQs covering: DBMS characteristics, ER diagrams, weak entities, relational model, relational algebra, SQL clauses, DML commands, functional dependencies, normal forms, ACID properties, 2PL, timestamp ordering]

Part – B: Descriptive (80 marks)

[8 questions on ER diagrams, relational algebra, SQL queries, normalization, FD closure, lossless decomposition, transactions, concurrency control]

3 Design and Analysis of Algorithms (CS3004-1)

3.1 MSE-I – February 2025

NMAM INSTITUTE OF TECHNOLOGY, NITTE

Off-Campus Centre of Nitte (Deemed to be University)

IV SEM B.Tech (CBCS) Mid Semester Examinations - I, February 2025

CS3004-1 – DESIGN AND ANALYSIS OF ALGORITHMS

(For AD, AM, CB, CC, CS, IS)

Duration: 1 Hour Max. Marks: 15

Part – A: MCQs (3 marks)

1. $T(n) = 7n^3 + 10n^2 + 4$ belongs to: (A) $O(n)$ (B) $O(n^2)$ (C) $O(n^3)$ (D) $O(n^4)$
2. Time complexity of $T(n) = 2T(n/2) + n$: (A) $O(n)$ (B) $O(n \log n)$ (C) $O(n^2)$ (D) $O(\log n)$
3. Minimum comparisons to find minimum in array of n elements: (A) n (B) $n - 1$ (C) $n + 1$ (D) $2n$

Part – B: Descriptive (12 marks)

1. a) Solve TSP using brute force for 4-vertex graph with edges: 1-2:10, 2-3:35, 3-4:25, 4-1:20, 1-3:15, 2-4:30. [6 marks]

   b) Quick Sort pseudocode. Trace for [38, 27, 43, 3, 9, 82, 10]. [6 marks]
2. a) Merge Sort algorithm with time complexity derivation. [6 marks]

   b) Tower of Hanoi: recursive algorithm and time complexity. [6 marks]

3.2 MSE-II – April 2025

NMAM INSTITUTE OF TECHNOLOGY, NITTE

Off-Campus Centre of Nitte (Deemed to be University)

IV SEM B.Tech (CBCS) Mid Semester Examinations - II, April 2025

CS3004-1 – DESIGN AND ANALYSIS OF ALGORITHMS

(For AD, AM, CB, CC, CS, IS)

Duration: 1 Hour Max. Marks: 20

Part – A: MCQs (4 marks)

1. Topological sorting requires: (A) Tree (B) DAG (C) Complete graph (D) Bipartite graph
2. DFS uses: (A) Queue (B) Heap (C) Priority queue (D) Stack
3. Distribution counting sort complexity: (A) $\Theta(n^2)$ (B) $O(n \log n)$ (C) $O(2^n)$ (D) $\Theta(n)$
4. Dynamic programming builds from: (A) Overlapping subproblems (B) Single recursive call (C) Greedy (D) Divide and conquer

Part – B: Descriptive (16 marks)

1. a) Insertion sort: Sort E, X, A, M, P, L, E alphabetically. [4 marks]

   b) Heap sort: Sort 90, 89, 56, 72, 67, 12, 21, 34, 45. [4 marks]
2. a) AVL tree construction for $\{2, 9, 3, 12, 15, 8, 11\}$. [4 marks]

   b) Topological sorting using Source Removal method. [4 marks]
3. a) Comparison Counting sort algorithm. Compare with Selection and Bubble sort. [4 marks]

   b) Horspool algorithm: Search TIME in "A STITCH IN TIME SAVES NINE". [4 marks]

3.3 SEE – May 2025

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)

Fourth Semester B.Tech (CBCS) Degree Examinations, May 2025
CS3004-1 – DESIGN AND ANALYSIS OF ALGORITHMS
(For CS, IS, AM, CC, AD, CB)

Duration: 3 Hours Max. Marks: 100

Part – A: MCQs (20 marks)

[20 MCQs on Quick sort, Binary search, Linear search, asymptotic analysis, recursion, Divide and Conquer, Dynamic programming, Floyd-Warshall, Horspool, Counting sort, Heap sort, AVL trees, B-trees, Branch and Bound, Backtracking, Dijkstra, Huffman codes]

Part – B: Descriptive (80 marks)

1. Performance Analysis, Selection sort [16 marks]
2. Non-recursive analysis, Merge sort [16 marks]
3. Heap construction, Horspool [16 marks]
4. B-Tree insertion, Topological sorting [16 marks]
5. Quick sort, Class P/NP/NP-Complete [16 marks]
6. Job Assignment (Branch and Bound), Subset Sum (Backtracking) [16 marks]

4 Innovation and Design Thinking (ME1654-1)

4.1 MSE-I – February 2025

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)

IV SEM B.Tech (CBCS) Mid Semester Examinations - I, February 2025
ME1654-1 – INNOVATION AND DESIGN THINKING
(For AD, AM, CB, CC, CS, IS)

Duration: 1 Hour Max. Marks: 15

Part – A: MCQs (3 marks)

1. Design Thinking focuses on: (A) Strict rules (B) Understanding user needs (C) Minimizing collaboration (D) Avoiding experimentation
2. "Empathy" means: (A) Finding user faults (B) Understanding users' feelings and motivations (C) Designing on assumptions (D) Testing without users
3. "Define" stage is about: (A) Generating ideas (B) Articulating problem from empathy insights (C) Building prototypes (D) Observing users

Part – B: Descriptive (12 marks)

1. a) Five stages of Design Thinking with examples. [6 marks]

   b) Importance of empathy. Methods to build empathy. [6 marks]

2. a) Ideation phase. Two ideation techniques. [6 marks]

   b) Prototyping: types of prototypes. [6 marks]

5 July 2023 & June 2022 MSE-II Papers

5.1 Linear Algebra (21CS401) – July 2023

[MCQs on dimension of symmetric matrices, linear transformations, basis, null space + Descriptive on basis for P_2, kernel/range, Gram-Schmidt, least squares]

5.2 DAA (20CS402) – July 2023

[MCQs on Heap Sort, AVL trees, DFS, Horspool + Descriptive on Heap Sort, AVL construction, DFS/BFS, Horspool string matching]

5.3 Microprocessor (21CS403) – July 2023

[MCQs on 8086 data bus, loop counter, physical address, flags + Descriptive on 8086 architecture, addressing modes, ALP for largest element, interrupt structure]

5.4 SEPM (21CS404) – July 2023

[MCQs on Waterfall model, SRS, non-functional requirements, Agile + Descriptive on Waterfall model, Agile vs Waterfall, SRS components, Use Case diagram]

5.5 Linear Algebra (20CS401) – June 2022

[MCQs on determinants, linear dependence, dimension, Rank-Nullity + Descriptive on subspaces, basis, linear transformations, orthonormal basis, projections]

5.6 DAA (21IS402) – June 2022

[MCQs on Quick Sort, B-tree, topological sorting, BFS + Descriptive on Quick Sort, B-tree construction, topological sorting, Horspool]

Part II
Descriptive Questions Only (From PDF Source)

6 Software Engineering & Project Management

6.1 CS2103-1 SEE – May 2025

1. a) Essential attributes of good software. Fundamental SE activities.
   b) Incremental Development with diagram.
2. a) ACM/IEEE Code of Ethics.
   b) Boehm's Spiral Model with diagram.
3. a) Structure of Requirements Document.
   b) Types of Non-Functional Requirements with diagram.
4. a) Requirements Elicitation and Analysis Process with diagram.
   b) Scenario for MHC-PMS medical history. Use cases.
5. a) System Testing.
   b) Five key UML diagram types with purposes.
6. a) Concept of system models.
   b) Language processing system architecture.
7. a) Four test strategies for software development.
   b) ATM System software architecture.
8. a) Project scheduling concept and significance.
   b) Configuration management concept.
9. a) Key stages in component measurement.
   b) Risk identification in project management.

6.2 CS2103-1 SEE – May 2024

1. a) Software development life cycle.
   b) Waterfall vs Spiral model.
2. a) V-model in plan-driven process.
   b) Boehm's spiral model.
3. a) Functional vs non-functional requirements.
   b) Incremental model advantages/disadvantages.
4. a) Data flow model (insulin pump example).
   b) Acceptance testing process stages.
5. a) Sequence models and Use-case models.
   b) Extreme programming and practices.
6. a) Weather Station object classes and diagrams.
   b) Agile methods and principles.
7. a) Risk and Risk management with categories.
   b) Project planning process.
8. a) Software review process phases.
   b) Software product metrics and types.

6.3 AD1103-1 SEE – May 2024

[Similar questions on software attributes, Waterfall model, SDLC, Scrum, requirements document, Agile principles, System Testing, activity diagrams, object classes, testing strategies, system models, architectural views, risk identification, configuration management, project scheduling, quality management]

6.4 AD1103-1 SEE – May 2025

[Questions on requirements document, requirement validation, interview types, SDLC phases, software attributes, Waterfall model, Scrum framework, System Testing, UML diagrams, system models, language processing, test strategies, ATM architecture, project scheduling, configuration management, measurement process, risk identification]

6.5 19IS8X83 SEE – May-June 2023

[Questions on Software Engineering definition, incremental development, requirements specification, software attributes, Waterfall model, requirement validation, design process, Boehm's spiral, state diagrams, object-oriented design, sequence diagrams, agile principles, MVC pattern, transaction processing, generalization, scrum process, risk management, communication factors, software pricing, project planning workflow, software review phases, cohesive groups]

6.6 CS2103-1 MSE-I – February 2025

[Questions on software attributes, Incremental Development, ACM/IEEE Code of Ethics, Spiral Model, Requirements Document structure, Non-Functional Requirements, Elicitation and Analysis, MHC-PMS scenarios]

6.7 CS2103-1 MSE-I – February 2024

[Questions on ethical issues, software applications, spiral model, software attributes, SRS document, incremental model, Waterfall vs incremental, non-functional requirements, software engineering discipline, system requirements, MHC-PMS, elicitation and analysis]

6.8 21CS404 MSE-I – June 2023

[Questions on software attributes, non-functional requirements, Waterfall model, application types, professional responsibility, Boehm's spiral, requirements document structure, elicitation process]

7 Database Management Systems (Additional)

7.1 CS2102-1 SEE – July 2024

[Questions on database characteristics, relational algebra operations, ER diagrams for flight management, grouping in relational algebra, aggregate functions, three-schema architecture, Sailors-Boats-Reserves queries, Employee-Works-Company-Manages queries, views, FD decomposition, lossless join, SQL constraints, order processing queries, normalization, 2PL protocol, indexing, interleaved execution anomalies, B+ tree insertion]

7.2 20CS502 SEE – January 2023

[Questions on database characteristics, flight management ER diagram, database design phases, structural constraints, recursive relationships, relational algebra queries, division, natural join, aggregate functions, Employee-Project queries, ER-to-Relational mapping, grouping, banking database SQL, SQL constraints, student enrollment queries, views, FD equivalence, informal design guidelines, normalization, minimal cover, 2PL, NoSQL aggregate models, interleaved execution, CAP theorem]

7.3 CS2102-1 MSE-I – February 2024

[Questions on Three Schema Architecture, mappings, ER diagram for Engineering College, database characteristics, domain and referential integrity, entity attributes, ER-to-relational conversion, good database design, characteristics of relations]

8 Design and Analysis of Algorithms (Additional)

8.1 CS3004-1 MSE-I – February 2024

[Questions on algorithm properties with linear search example, function value changes (log, linear, quadratic, exponential, sqrt), algorithm analysis framework, asymptotic notations, recursive factorial analysis, brute force sorting, Merge sort with analysis, brute force string matching]

8.2 21CS402 MSE-I – June 2023

[Questions on algorithm design and analysis process flow, recursive algorithm for 2^n, asymptotic notations with proofs, sequential search complexity, brute force string matching, Merge sort with alphabetical sorting, Quick sort with analysis, exhaustive search for knapsack]

8.3 21CS402 Makeup SEE – September 2023

[Questions on distinct elements algorithm, brute force string matching, Merge sort with worst case, sequential search, selection sort complexity, Quick sort, Big Oh and Big Theta, binary digits algorithm, Divide and Conquer, Binary search]

9 Linear Algebra (Additional)

9.1 MA2005-1 MSE-I – February 2024

[Questions on Gauss-Seidel with 4 iterations, power method, LU factorization, Cramers rule, augmented matrix solutions, eigenvalues and eigenvectors, eigenvalue properties, iterative methods conditions, determinant evaluation, matrix multiplications, echelon form, row and column space basis]

9.2 21CS401 MSE-II – July 2023

[Questions on dimension of symmetric matrices, linear transformations (one-to-one, onto), basis properties, null space subspace, basis for P_2, kernel and range, Gram-Schmidt orthonormalization, least squares solution]

9.3 21CS401 MSE-I – June 2023

[Questions on eigenvalues, echelon form, homogeneous systems, consistent systems, Gauss elimination, LU factorization, power method, matrix diagonalization]

9.4 MA2006-1 MSE-I – February 2024

[Questions on power method convergence, LU decomposition, Gauss-Seidel convergence, eigenvalue problems, matrix operations]

9.5 19MA8X02 SEE – January 2023

[Questions on determinants, matrix inverse, linear independence, eigenvalues, eigenvectors, Cayley-Hamilton theorem, diagonalization, Jordan form, inner product spaces, Gram-Schmidt, QR factorization]

9.6 20IS401 Supplementary – September 2022

[Questions on linear systems, echelon form, LU factorization, iterative methods, eigenvalues, power method, diagonalization, orthogonalization]

10 Microprocessor and Embedded Systems

10.1 CS3005-1 SEE – May 2025

[Questions on 8086 architecture, memory segmentation, addressing modes, instruction set, ALP for string operations, array processing, arithmetic operations, interrupt handling, 8255 PPI, timer/counter operations]

10.2 CS3005-1 SEE – May 2024

[Questions on microprocessor evolution, 8086 pin diagram, bus interface unit, execution unit, memory organization, stack operations, procedure calls, interrupts, DMA, serial communication]

10.3 CS3005-1 SEE – July 2024

[Questions on 8086 internal architecture, flag register, addressing modes, instruction formats, ALP programs, macros, procedures, interrupt service routines, peripheral interfacing]

10.4 CS3005-1 MSE-I – February 2024

[Questions on microprocessor basics, 8086 architecture, memory segmentation, instruction execution, addressing modes, data transfer instructions]

10.5 21CS403 MSE-I – June 2023

[Questions on 8086 data bus width, registers, physical address calculation, flags, architecture, addressing modes with examples]

10.6 21CS403 SEE – August 2023

[Questions on 8086 comprehensive coverage, ALPs, interrupt handling, I/O interfacing, timer applications]

10.7 19EC503 Supplementary – September 2022

[Questions on microprocessor fundamentals, 8086 programming, peripheral interfacing]

10.8 20EC503 SEE – January 2023

[Questions on 8086 architecture, instruction set, assembly language programming, I/O interfacing]

10.9 20IS501 SEE – January 2023

[Questions on microprocessor systems, 8086 programming, interrupt systems, peripheral devices]

11 Computer Organization and Architecture

11.1 CS2101-1 SEE – July 2024

[Questions on computer organization fundamentals, instruction set architecture, CPU design, memory hierarchy, cache organization, pipelining, I/O systems, parallel processing]

12 Innovation and Design Thinking (Additional)

12.1 ME1654-1 SEE – May 2025

[Questions on Design Thinking process, Empathize stage methods, Define stage techniques, Ideate stage tools, Prototype types, Test stage approaches, innovation frameworks, creativity techniques, user research methods, design challenges]

13 Supplementary Papers

13.1 19CS502 DBMS Supplementary – September 2022

[Questions on database concepts, ER modeling, relational algebra, SQL, normalization, transaction management]

13.2 20IS402 DAA Supplementary – September 2022

[Questions on algorithm analysis, sorting algorithms, searching algorithms, graph algorithms, dynamic programming, greedy algorithms]

— End of Comprehensive Question Bank —

This document contains ALL examination papers from:
SEE (May 2025, May 2024, July 2024, January 2023, August 2023)
MSE-I (February 2025, February 2024, June 2023)
MSE-II (April 2025, July 2023)
Supplementary (September 2022)

Subjects: Linear Algebra, DBMS, DAA, SEPM, Design Thinking, Microprocessor, Computer Organization